

Question Two– Use Python

- Generate appropriate graphs to present the data
- Perform descriptive analysis of the dataset
- Formulate five objectives, perform tests of hypothesis and discuss the results

```
In [57]: # necessary libraries
import pandas as pd # data transformation
import matplotlib.pyplot as plt # data visualization
import seaborn as sns # data visualization
import pandoc
```

```
In [61]: # Load the data
data = pd.read_csv("C:\\Users\\franc\\Datascience_python\\dataset\\Housing.csv")

# Transform the data into a pandas DataFrame (df)
df = pd.DataFrame(data)
print(df.head())
```

	price	area	bedrooms	bathrooms	stories	mainroad	guestroom	basement	\
0	13.300	7.42	4	2	3	yes	no	no	
1	12.250	8.96	4	4	4	yes	no	no	
2	12.250	9.96	3	2	2	yes	no	yes	
3	12.215	7.50	4	2	2	yes	no	yes	
4	11.410	7.42	4	1	2	yes	yes	yes	

	hotwaterheating	airconditioning	parking	prefarea	furnishingstatus
0	no	yes	2	yes	furnished
1	no	yes	3	no	furnished
2	no	no	2	yes	semi-furnished
3	no	yes	3	yes	furnished
4	no	yes	2	no	furnished

```
In [63]: #checking duplicate values
df.nunique()
```

```
Out[63]: price          219
         area           284
         bedrooms       6
         bathrooms      4
         stories        4
         mainroad       2
         guestroom      2
         basement       2
         hotwaterheating 2
         airconditioning 2
         parking        4
         prefarea       2
         furnishingstatus 3
         dtype: int64
```

```
In [65]: # check unique values
print(df['mainroad'].unique())
"""
print(df['guestroom'].unique())
print(df['basement'].unique())
print(df['hotwaterheating'].unique())
print(df['airconditioning'].unique())
print(df['prefarea'].unique())
"""
print(df['furnishingstatus'].unique())
```

```
['yes' 'no']
['furnished' 'semi-furnished' 'unfurnished']
```

Convert categorical variables to numerical

```
In [68]: # Convert categorical variables to numerical
df_encoded = df.copy()

# Identify categorical columns
categorical_columns = df_encoded.select_dtypes(include=['object']).columns

# Convert 'yes'/'no' to 1/0
binary_columns = ['mainroad', 'guestroom', 'basement', 'hotwaterheating', 'aircondi
for col in binary_columns:
    df_encoded[col] = df_encoded[col].map({'yes': 1, 'no': 0})

# Convert 'furnishingstatus' to numerical encoding
df_encoded['furnishingstatus'] = df_encoded['furnishingstatus'].map({'furnished': 2

print(df_encoded.head())
```

	price	area	bedrooms	bathrooms	stories	mainroad	guestroom	basement	\
0	13.300	7.42	4	2	3	1	0	0	
1	12.250	8.96	4	4	4	1	0	0	
2	12.250	9.96	3	2	2	1	0	1	
3	12.215	7.50	4	2	2	1	0	1	
4	11.410	7.42	4	1	2	1	1	1	

	hotwaterheating	airconditioning	parking	prefarea	furnishingstatus
0	0	1	2	1	2
1	0	1	3	0	2
2	0	0	2	1	1
3	0	1	3	1	2
4	0	1	2	0	2

a) appropriate graphs to present the data

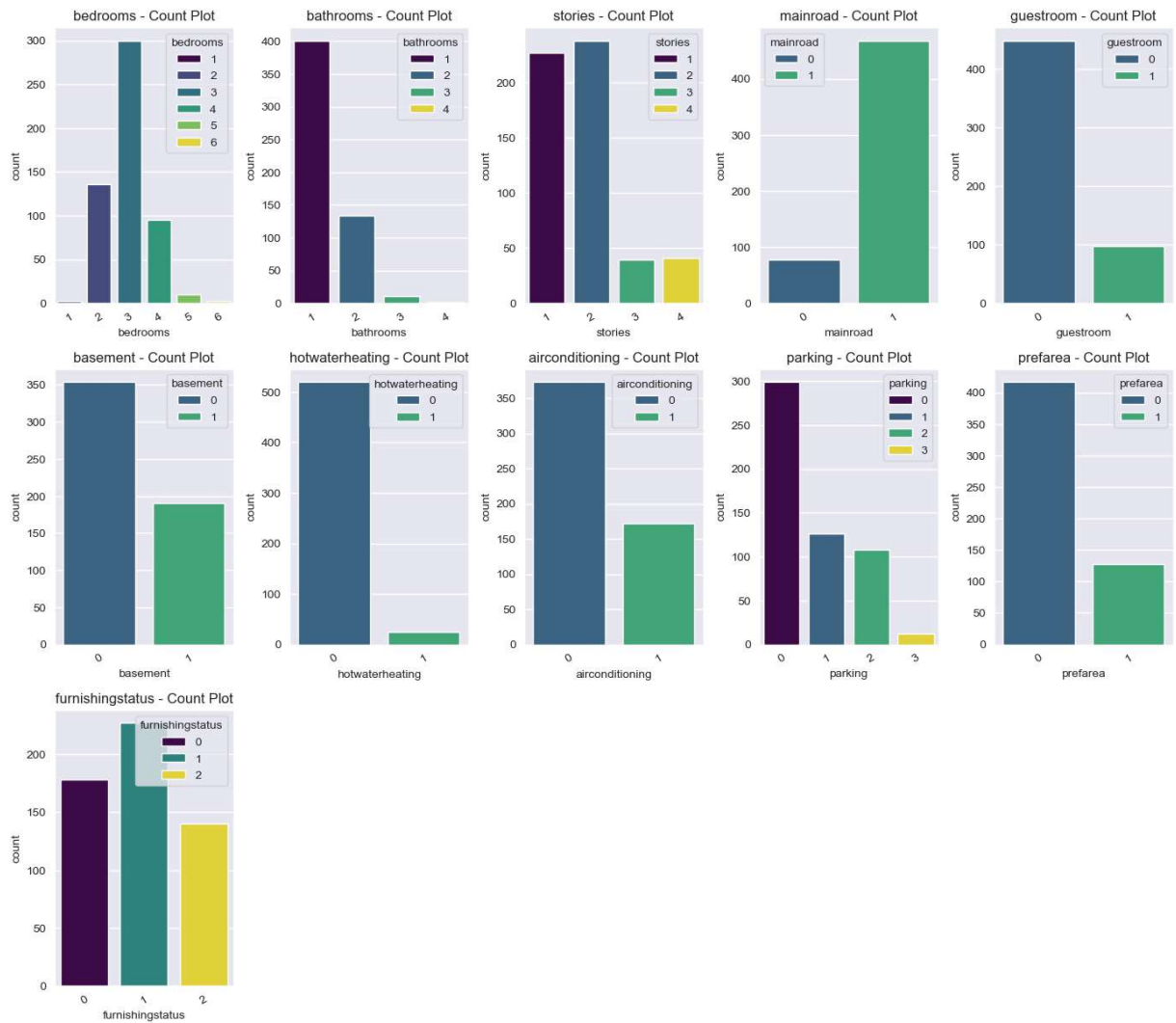
```
In [71]: # count plot for the data
sns.set_style("darkgrid")

# Select columns
columns = ['bedrooms', 'bathrooms', 'stories', 'mainroad',
           'guestroom', 'basement', 'hotwaterheating', 'airconditioning',
           'parking', 'prefarea', 'furnishingstatus']

# Define figure size
plt.figure(figsize=(14, len(columns) * 4))

# Loop through columns and create count plots
for index, feature in enumerate(columns, start=1):
    plt.subplot(len(columns), 5, index)
    sns.countplot(x=df_encoded[feature], hue=df_encoded[feature], palette="viridis")
    plt.title(f"{feature} - Count Plot")
    plt.xticks(rotation=30)

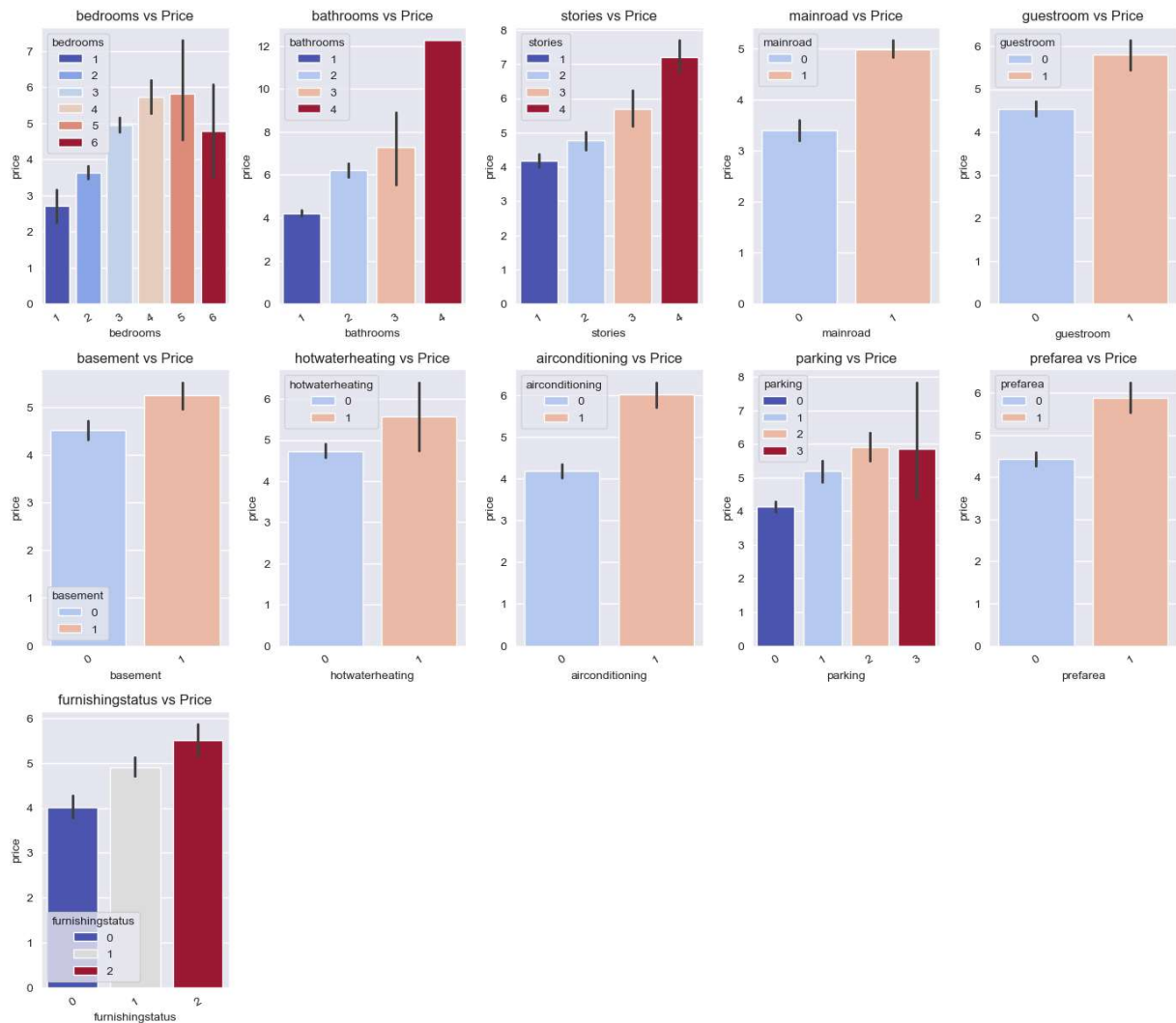
plt.tight_layout()
plt.show()
```



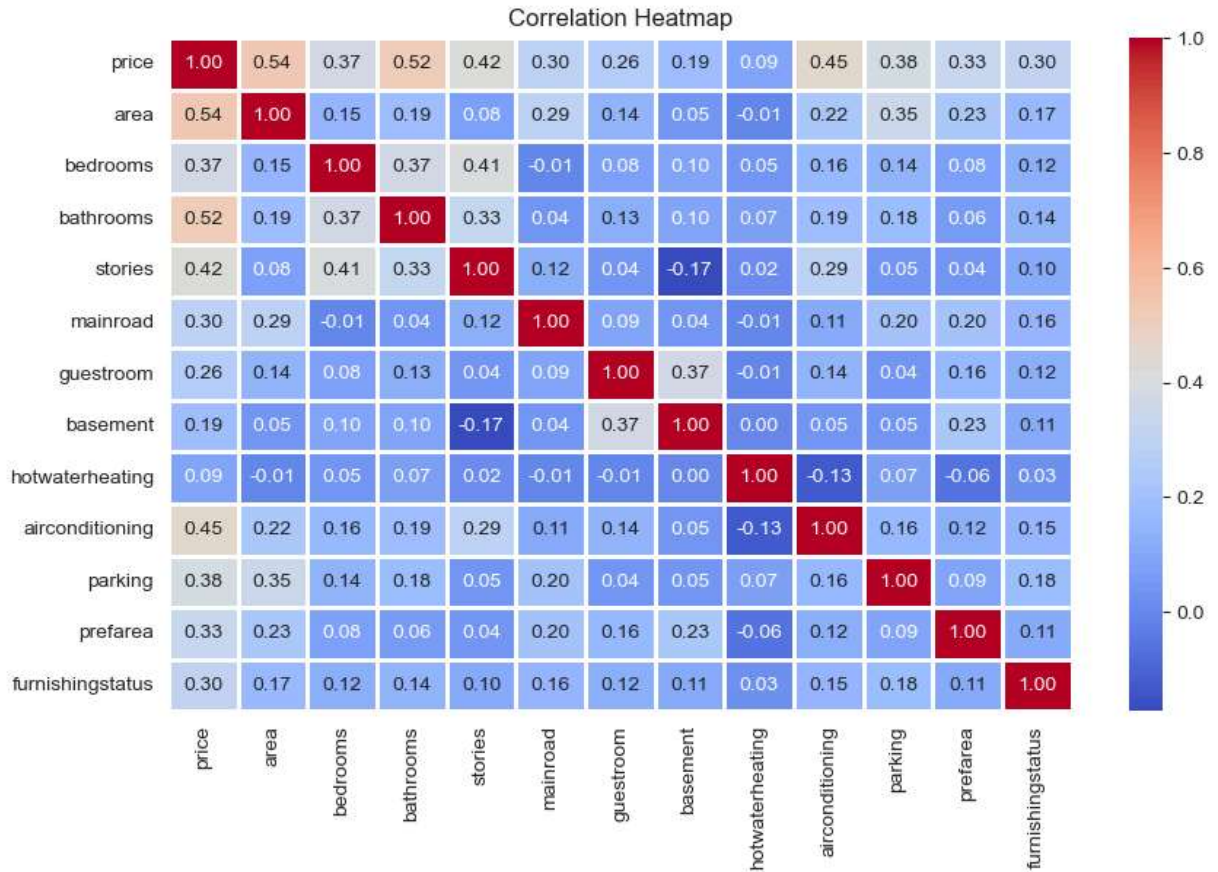
```
In [72]: # how columns variables impact house price:
plt.figure(figsize=(14, len(columns) * 4))

for index, feature in enumerate(columns, start=1):
    plt.subplot(len(columns), 5, index)
    sns.barplot(x=df_encoded[feature], y=df_encoded['price'], hue=df_encoded[feature])
    plt.title(f"{feature} vs Price")
    plt.xticks(rotation=30)

plt.tight_layout()
plt.show()
```



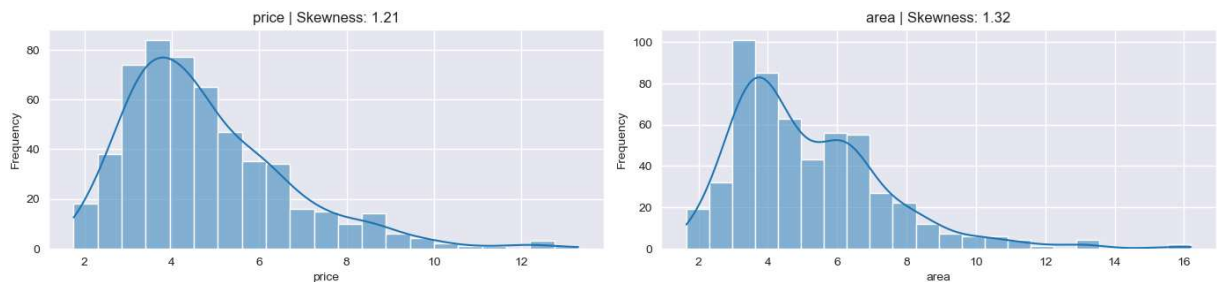
```
In [74]: # Generate the heatmap
plt.figure(figsize=(10, 6))
sns.heatmap(df_encoded.corr(), annot=True, fmt = '.2f' , cmap='coolwarm', linewidths=
plt.title('Correlation Heatmap')
plt.show()
```



```
In [79]: # Distribution of price and area
sns.set_style("darkgrid")

columns = ['price', 'area']
plt.figure(figsize=(14, len(columns) * 3))
for idx, feature in enumerate(columns,1):
    plt.subplot(len(columns),2,idx)
    sns.histplot(df_encoded[feature],kde = True)
    plt.ylabel('Frequency')
    plt.title(f"{feature} | Skewness: {round(df_encoded[feature].skew(),2)}")

plt.tight_layout()
plt.show()
```



```
In [83]: # Scatter plot of Price vs Area
plt.figure(figsize=(5, 5))
sns.scatterplot(x=df_encoded['area'], y=df_encoded['price'])
plt.title('Price vs Area')
plt.xlabel('Area')
```

```
plt.ylabel('Price')  
plt.show()
```



b) Perform descriptive analysis of the dataset

```
In [87]: # Descriptive statistics  
df_encoded.describe()
```

Out[87]:

	price	area	bedrooms	bathrooms	stories	mainroad	guestroom
count	545.000000	545.000000	545.000000	545.000000	545.000000	545.000000	545.000000
mean	4.766729	5.150541	2.965138	1.286239	1.805505	0.858716	0.177982
std	1.870440	2.170141	0.738064	0.502470	0.867492	0.348635	0.382849
min	1.750000	1.650000	1.000000	1.000000	1.000000	0.000000	0.000000
25%	3.430000	3.600000	2.000000	1.000000	1.000000	1.000000	0.000000
50%	4.340000	4.600000	3.000000	1.000000	2.000000	1.000000	0.000000
75%	5.740000	6.360000	3.000000	2.000000	2.000000	1.000000	0.000000
max	13.300000	16.200000	6.000000	4.000000	4.000000	1.000000	1.000000

```

In [85]: """Box Plot for examining the relationship price and Features"""

sns.set_style("darkgrid")

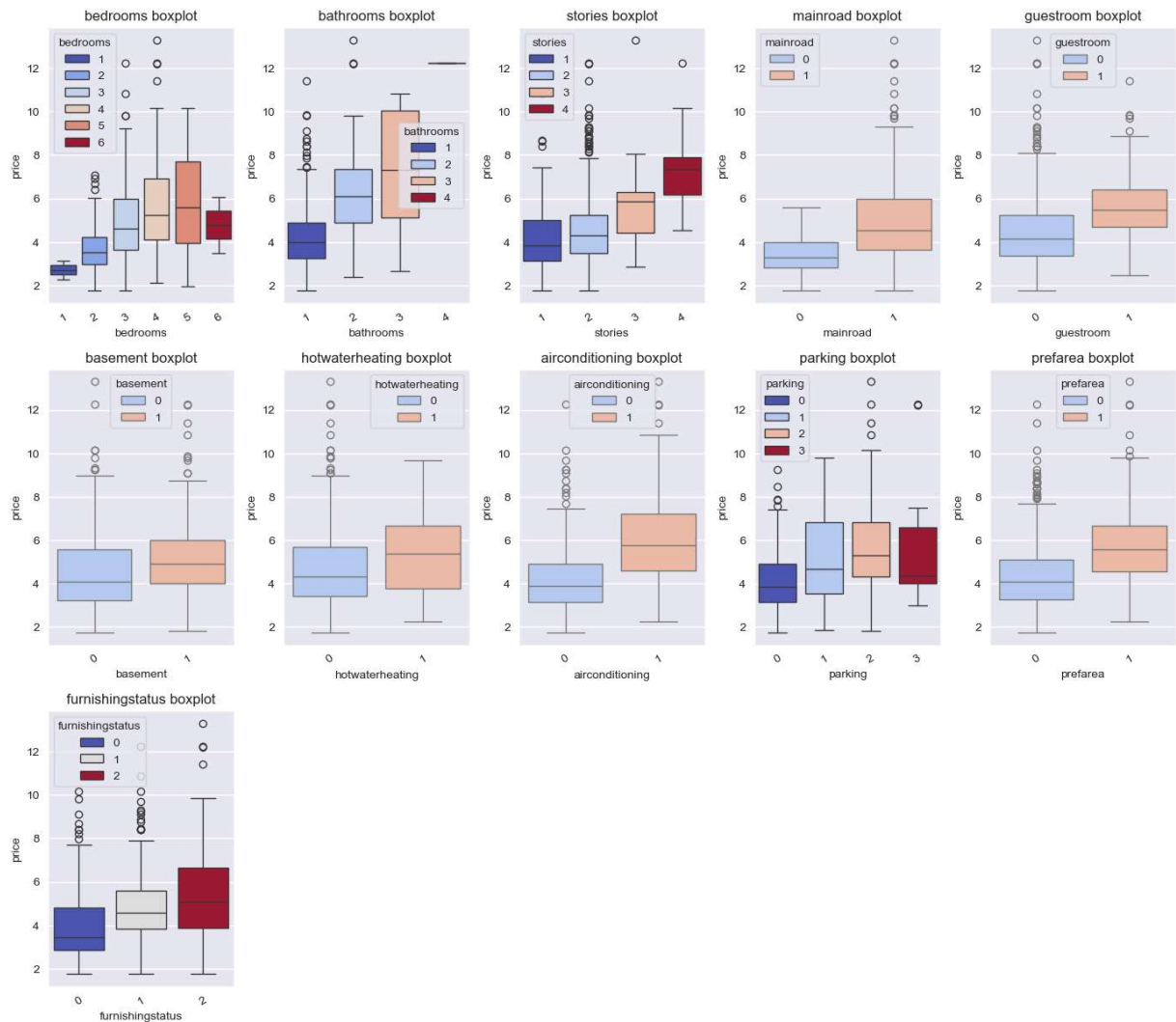
# Select columns
columns = ['bedrooms', 'bathrooms', 'stories', 'mainroad',
           'guestroom', 'basement', 'hotwaterheating', 'airconditioning',
           'parking', 'prefarea', 'furnishingstatus']

plt.figure(figsize=(14, len(columns) * 4))

for index, feature in enumerate(columns, start=1):
    plt.subplot(len(columns), 5, index)
    sns.boxplot(x=df_encoded[feature], y=df_encoded['price'], data=df_encoded, hue=
    plt.title(f"{feature} boxplot")
    plt.xticks(rotation=30)

plt.tight_layout()
plt.show()

```

c) Formulate Five Objectives and Perform Hypothesis Testing

```
In [89]: # necessary libraries
import numpy as np
from scipy import stats
```

Objective 1: Does area affect price?

```
In [92]: # Define hypothesis
null_hypothesis = "Area has no effect on price."
alternate_hypothesis = "Area has an effect on price."
alpha = 0.05

# Perform Pearson correlation test
corr, p_value = stats.pearsonr(df_encoded['area'], df_encoded['price'])

# condition
if p_value <= alpha:
```

```

decision = "Reject"
conclusion = "There is sufficient evidence to claim area has an effect on price"
else:
    decision = "Fail to reject"
    conclusion = "There is statistically significant evidence that area has no effect on price"

# results
print(f"Correlation: {corr:.2f}, P-value: {p_value}")
print(f"Decision: {decision} the null hypothesis at alpha={alpha}.")
print("Conclusion:", conclusion)

```

Correlation: 0.54, P-value: 7.388225401521874e-42

Decision: Reject the null hypothesis at alpha=0.05.

Conclusion: There is sufficient evidence to claim area has an effect on price.

Objective 2: Do houses near the main road have a higher price?

mainroad is a categorical variable (yes/no mapped to 1/0),

we compare the means of house prices for both groups.

```

In [95]: # Define hypothesis
null_hypothesis = "Main road has no effect on price."
alternate_hypothesis = "Main road has an effect on price."
alpha = 0.05

# Perform t-test
t_stat, p_value = stats.ttest_ind(df_encoded['mainroad'], df_encoded['price'])

# condition
if p_value <= alpha:
    decision = "Reject"
    conclusion = "There is sufficient evidence to claim main road has an effect on price"
else:
    decision = "Fail to reject"
    conclusion = "There is statistically significant evidence that main road has no effect on price"

# results
print(f"T-statistic: {t_stat:.4f}, P-value: {p_value}")
print(f"Decision: {decision} the null hypothesis at alpha={alpha}.")
print("Conclusion:", conclusion)

```

T-statistic: -47.9507, P-value: 1.4307547292241981e-270

Decision: Reject the null hypothesis at alpha=0.05.

Conclusion: There is sufficient evidence to claim main road has an effect on price.

Objective 3: Does the number of bedrooms impact price?

I am using an Ordinary Least Squares (OLS) regression to check if bedrooms significantly affect price.

```
In [98]: # neccesary libraries
import statsmodels.api as sm
```

```
In [100... # Define hypothesis
null_hypothesis = "The number of bedrooms has no significant impact on the price."
alternate_hypothesis = "The number of bedrooms has significant impact on the price."

# Define independent (X) and dependent (y) variables
X = df_encoded[['bedrooms']]
y = df_encoded['price']

# Add a constant for intercept
X = sm.add_constant(X)

# Fit the model
model = sm.OLS(y, X).fit()

# Display summary
print(model.summary());

# results
print("Decision: Reject' the null hypothesis at alpha 0.05.")
print("Conclusion: There is sufficient evidence to claim The number of bedrooms ha
```

OLS Regression Results

```
=====
Dep. Variable:          price    R-squared:                0.134
Model:                  OLS      Adj. R-squared:           0.133
Method:                 Least Squares    F-statistic:         84.25
Date:                  Mon, 10 Mar 2025    Prob (F-statistic):    9.08e-19
Time:                  01:35:02    Log-Likelihood:       -1074.8
No. Observations:      545    AIC:                  2154.
Df Residuals:          543    BIC:                  2162.
Df Model:               1
Covariance Type:       nonrobust
=====
```

	coef	std err	t	P> t	[0.025	0.975]
const	2.0127	0.309	6.510	0.000	1.405	2.620
bedrooms	0.9288	0.101	9.179	0.000	0.730	1.128

```
=====
Omnibus:                91.744    Durbin-Watson:         0.261
Prob(Omnibus):           0.000    Jarque-Bera (JB):      162.482
Skew:                    0.997    Prob(JB):              5.22e-36
Kurtosis:                4.784    Cond. No.              13.9
=====
```

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Decision: Reject' the null hypothesis at alpha 0.05.

Conclusion: There is sufficient evidence to claim The number of bedrooms has an effect on price.

Objective 4: Is there a significant price difference between furnished and unfurnished houses?

Since furnishingstatus has three categories (furnished, semi-furnished, unfurnished),

I will use ANOVA (Analysis of Variance).

(One-Way ANOVA Test)

```
In [103... # Define hypothesis
null_hypothesis = "There is no significant price difference between furnishing stat
alternate_hypothesis = "There is a significant price difference between furnished,

# Extract price data for each furnishing category
furnished = df_encoded[df_encoded['furnishingstatus'] == 2]['price']
semi_furnished = df_encoded[df_encoded['furnishingstatus'] == 1]['price']
unfurnished = df_encoded[df_encoded['furnishingstatus'] == 0]['price']

# Perform ANOVA test
f_stat, p_value = stats.f_oneway(furnished, semi_furnished, unfurnished)

# Decision
alpha = 0.05
if p_value <= alpha:
    decision = "Reject"
    conclusion = "There is a significant difference in price among furnishing statu
else:
    decision = "Fail to reject"
    conclusion = "There is no significant difference in price among furnishing stat

print(f"F-statistic: {f_stat:.4f}, P-value: {p_value}")
print(f"Decision: {decision} the null hypothesis at alpha={alpha}.")
print("Conclusion:", conclusion)
```

F-statistic: 28.2721, P-value: 2.0928668857953132e-12

Decision: Reject the null hypothesis at alpha=0.05.

Conclusion: There is a significant difference in price among furnishing statuses.

Objective 5: Does having air conditioning increase the price?

```
In [106... # Define hypothesis
null_hypothesis = "Houses with air conditioning have the same average price as thos
alternate_hypothesis = "Houses with air conditioning have a higher price."

# Perform t-test
t_stat, p_value = stats.ttest_ind(df_encoded['airconditioning'], df_encoded['price'])

# Decision
alpha = 0.05
if p_value <= alpha:
```

```
decision = "Reject"
conclusion = "There is sufficient evidence to suggest that air-conditioned hous
else:
decision = "Fail to reject"
conclusion = "There is no significant evidence that air-conditioned houses have

print(f"T-statistic: {t_stat:.4f}, P-value: {p_value}")
print(f"Decision: {decision} the null hypothesis at alpha={alpha}.")
print("Conclusion:", conclusion)
```

T-statistic: -53.9130, P-value: 1.502079436517493e-309

Decision: Reject the null hypothesis at alpha=0.05.

Conclusion: There is sufficient evidence to suggest that air-conditioned houses have a different price.